INCH - POUND

MIL-DTL-5882E
17 September 2013
Superseding
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DETAIL SPECIFICATION

TRANSMITTER, PRESSURE, MULTIPURPOSE, 0-50 PSI, MS28005-1

This specification is approved for use by Department of the Air Force and is available for use by all Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating, 320 degrees movement synchro style, general purpose pressure gage transmitter.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5350	-	Transmitters, Synchro Operated, Aircraft General Specification For
MIL-I-7085	-	Indicator, Pressure, Multipurpose, 0-50 Psi
MIL-B-27497	-	Bearings, Jewel, Sapphire, Synthetic

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MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-810	-	Environmental Test Methods & Engineering Guidelines
MS28005	-	Transmitter, Pressure-Synchro, Aircraft
MS28010	-	Indicators, Pressure Synchro, Single, 2 – Inch Size

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	-	Reliability Testing For Engineering Development, Qualification & Production

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Qualification. The transmitter furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (See 4.1 and 6.3).

3.2 General requirements. The requirements of MlL-PRF-5350 are applicable as requirements of this specification. Additional requirements shall be as specified herein. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2.2 Reclaimed materials. Reclaimed materials shall be utilized to the maximum extent possible within the quality limits required by this document.

3.3. Design and construction.

3.3.1 General Design. The transmitters shall be designed for use with an MS28010-1 indicator conforming to MIL-I-7085.

3.3.2 Hermetically sealed case. The case shall provide a hermetically sealed enclosure for the entire internal mechanism of the transmitter. The design of the case shall permit removal and replacement of the internal mechanism and resealing of the case. This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any environment to which the transmitter may be subjected.

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3.3.3 Outline dimensions. The outline dimensions of the transmitter shall conform to MS28005.

3.3.4 Bosses. The bosses, as shown on MS28005, shall be provided for external pressure and vent connections to the pressure sensitive element and the opposing pressure element, respectively. The bosses shall be identified with white lettering "P" and "V" permanently inscribed 0.125 inch in height.

3.3.5 Operation. The transmitter shall be capable of operating in any position.

3.3.6 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the pointer of an. indicator to travel beyond full scale to a point approximately midway between the last and the zero graduation.

3.3.7 Pressure connections. The pressure transmitter shall have two pressure connections (external pressure connection and vent connection). The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the external pressure connection and the pressure applied to the vent connection, and the synchro signal output shall represent external pressure minus vent pressure. The angular position of the rotor of the synchro relative to the stator shall be essentially a linear function of external pressure minus vent pressure.

3.3.8 Silicone oil. The transmitter shall not contain silicone oil.

3.4 Bearings synthetic. When sapphire or ruby jewel synthetic bearings are used they shall be in accordance with MIL-B-27497.

3.5 Weight. The weight of the transmitter shall not exceed 1.4 pounds.

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. VERIFICATION

4.1 Classification of inspections. The inspection and testing of transmitter shall be classified as follows:

a. Qualification inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 Qualification inspection. Qualification tests shall consists of all tests under paragraph 4.4, except 4.4.24 of MIL-PRF-5350, and the tests of this specification.

4.3 Conformance inspection. Inspection shall consist of the following:

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a. Individual inspection (see 4.3.1)
b. Sampling inspection (see 4.3.2)

4.3.1 Individual inspections. Each transmitter submitted for acceptance under contract shall be subjected to the individual inspections specified in MIL-PRF-5350 and all tests specified in 4.5.

TABLE I. Multi-purpose pressure transmitter scale error tolerances, ± psi.

TEST CONDITION	TEST POINTS (PSI)						
	0	10	20	25	30	40	50
Room Temperature	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Low Temperature Operational	1.25	1.25	1.25	1.25	1.25	1.50	1.50
High Temperature Operational	1.25	1.25	1.25	1.25	1.25	1.50	1.50
Vibration (Post Test Operation)	1.00	1.00	1.00	1.00	1.00	1.00	1.25
Low Temperature Exposure (Operational)	1.25	1.25	1.25	1.25	1.25	1.50	1.50
Room Temp. following Low Temp. Exposure*	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Room Temp. following High Temp. Exposure*	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Seasoning	1.00	1.00	1.00	1.00	1.00	1.00	1.25
Endurance	1.25	1.25	1.25	1.25	1.25	1.50	1.50
Power Supply Variation*	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Salt Spray *	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Suction and Overpressure	1.50	1.50	1.50	1.50	1.50	1.50	1.50
Shock (Post Test Operation) *	0.75	0.75	0.75	0.75	0.75	0.75	0.75

*NOTE: The tolerances shown are applicable unless allowable shifts in calibration have occurred during preceding tests, in which case the tolerance of such preceding test shall apply.

4.3.2 Sampling. Sampling shall be as specified in MIL-PRF-5350.

4.4 Inspection conditions. Unless otherwise specified, the transmitter shall be tested in a horizontal position with the pressure connection at the top.

4.5 Test Methods.

4.5.1 Scale error. The test points and tolerances for the error test shall be as specified in Table I.

4.5.2 Friction. Tolerances for the friction test shall be as specified in Table II.

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TABLE II. Pressure transmitter.

Test	Tolerance ± psi
Vibration:	
Pointer oscillation	0.50
Friction:	
Room temperature	0.50
High temperature	0.75
Low temperature	1.00
Position error at 30 psi	0.50
Leak	0.10

4.5.3 Sealing. Pressure and vent connections shall be closed with suitable plugs for this test. Each transmitter shall be immersed in a suitable deaerated liquid, such as water, within an enclosure. The absolute pressure of the air above the liquid shall be reduced to 2.5 inches Hg. Any bubbles coming from within the transmitter case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air in the various exterior parts of the case shall not be considered as a leak. A wetting agent shall be used in the liquid to facilitate removal of surface bubbles.

4.5.3.1 The sealing test of MIL-PRF-5350 may be used as an alternate method.

4.5.4 Leak. A pressure of 50 pounds per square inch shall be applied to the pressure connection and a pressure of 20 pounds per square inch shall be applied to the vent connection. The connecting tubes shall be completely sealed at a point within 2 inches of each connection. During a period of one minute, the pointer shall not change its position by more than the tolerance specified in Table II. This test may be combined with that of the scale error test.

4.5.5 Damping. A pressure sufficient to produce full scale test pointer deflection shall be applied to the transmitter. A quick release valve shall be used to release the pressure. The time interval for the change from 50 psi to 5 psi indication shall be 9 ± 3 seconds. The test shall be conducted with lubricating oil with a viscosity of approximately 2,500 centistokes, such as grade 1005 oil at -65°F, or equivalent. The test shall be conducted at 77°F.

4.5.6 Suction and overpressure. The pressure port of the transmitter shall be subjected to a pressure of 5 ± 0.5 inches Hg absolute for a period of one minute. The pressure port shall then be subjected to a pressure of 100 psi for a period of ten minutes. Not less than 1 hour following the overpressure, the transmitter shall be tested for scale errors as specified in the individual tests on increasing pressures only. The scale error shall not exceed the tolerances specified in Table I.

4.5.7 Salt fog. After exposure to the specified salt fog test the transmitter shall be examined for corrosion and subjected to the sealing test. No scale or friction error test shall be required.

4.5.8 Power supply variation. Power supply variation tests shall allow fluctuations of ± volts and ± 20 percent of the nominal frequency.

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4.5.9 Reliability tests. The reliability test units shall be subjected to the following combined environmental conditions:

a. Temperature: -55°C to +121°C or remotely mounted transmitters or -55°C to +190°C for direct engine mounted transmitters.

b. Temperature cycling: Each temperature cycle shall be 8 hours in length. Following a room temperature dwell of 30 minutes, the chamber temperature will be dropped over a period of 15 minutes to -55°C. Following a 3 hour, -55°C soak, the chamber temperature shall be raised over a 45 minute period to the upper temperature range (+121°C or +190°C). The transmitter shall remain at the high temperature over a period of 3 hours after which the temperature shall be lowered to room temperature over a period of 30 minutes. This constitutes on cycle. This cycle shall continue throughout the duration of the reliability tests, except for moments of equipment shut down to assess transmitter performance.

c. Transmitter on-off cycling: The transmitters shall be off for 15 minutes out of each high temperature and low temperature soak as well as during the 30 minute room temperature period. The transmitters shall be on at all other times.

d. Pressure oscillations: The transmitters shall be subjected to input line pressure oscillations at a frequency of 1 Hz from 20 ± 1 to 30 ± 1 psi. Pressure oscillations shall occur whenever the transmitters are powered on. When the transmitters are powered off, the input line pressure shall be 0 psi.

e. Vibration: Throughout the pressure cycling, for 10 minutes of each hour the transmitters shall be subjected to sinusoidal vibration cycling in accordance with MIL-STD-8l0C, method 514.2, using curve D for shock mounted transmitters or curve L for direct engine mounted transmitters. At the end of 4 test cycles (32 hours) during the room temperature phase, reorient the transmitters to another axis and continue testing.

f. Input voltage: Nominal specified voltage.

At the end of each 4 cycles, when the transmitters are being reoriented on the vibration stand, the transmitters shall be scale error tested. Any transmitter which does not stay within the scale error tolerances for endurance in Table I shall be considered a failure. No preventative maintenance shall be performed. Lower test MTBF (O) shall not be less than 5000 hours. Accept - reject criteria shall be in accordance with MIL-HDBK-78l, (for guidance only) test plan I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities with the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the

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managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The multipurpose pressure transmitter covered by this specification is intended for use to transmit the indicated pressure in the fuel lubricating oil, and water injection systems of aircraft engines to an applicable indicator.

6.1.1 General. The instruction contained in section 6 of MIL-PRF-5350 are applicable to and form a part of this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Packaging (see 5.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-5882 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/ENSDAA (AF-71), 3001 Staff Drive, Suite 1AB81A, Tinker AFB, OK 73145 or email tinker.dsp@tinker.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

320 degrees
Remote, indicating
Synchro, style

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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